Template Document

Academic Support Agent

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Abstract

This guide is for authors who are preparing papers for *Network Science* using the LaTeX 2_ε document-preparation system and the Network Science class file (`nws.cls`).

1 Introduction

Newtork Science accepts submissions either in Microsoft Word or in LaTeX format. This guide describes the use and features of the class for the journal. It is based on the `article` class as discussed in the LaTeX manual (2nd edition) [**?**]. Commands which differ from the standard LaTeX 2_ε interface, or which are provided in addition to the standard interface, are explained in this guide (which is *not* a substitute for the LaTeX 2_ε manual itself). We assume you are already familiar with LaTeX.

Note that this guide uses the public `mathptmx` fonts. If you use this guide as a template please note that the final printed version of papers may use a different typeface so line and page breaks will change. Therefore do not put hard page or line breaks into the document

Authors planning to submit their papers in LaTeX 2_ε are advised to use `nws.cls` as early as possible in the creation of their files.

1.1 Introduction to LaTeX

LaTeX is constructed as a series of macros on top of the TeX typesetting program. LaTeX adds to TeX a collection of facilities which simplify typesetting for authors by allowing them to concentrate on the logical structure of the document rather than its visual layout. Careful use of the LaTeX mark-up philosophy results in uniform layout rather than the *ad hoc* results of some word-processing systems. Authors are advised to let the defaults control font selection etc., rather than tinker themselves.

The LaTeX system provides a consistent and comprehensive document preparation interface. Among other things, LaTeX can automatically number list entries, equations, figures, tables and footnotes, as well as sections and subsections. Using this numbering system, bibliographic citations, page references and cross references to any other numbered entity (e.g. sections, equations, figures) are straightforward.

1.2 The NWS document class

The use of document classes allows a simple change of style (or style option) to transform the appearance of your document. The NWS class preserves the standard LaTeX interface such that any document which can be produced using the standard LaTeX `article` class can also be produced with the NWS class.

1.3 General style issues

Use of LaTeX defaults will result in a pleasing uniformity of layout and font selection. Authors should resist the temptation to make *ad hoc* changes to these. Also avoid use of direct formatting unless really necessary. Papers will be edited as usual, and this process may be obstructed by the use of inserted line breaks, etc.

For general style issues, authors are referred to the 'Preparation of manuscripts' in the back cover of the journal. Authors who are interested in the details of style are referred to [**?**] and [**?**]. The language used in the journal is US English, and spelling should conform to this.

Use should be made of symbolic references (`\ref`) in order to protect against late changes of order, etc.

1.4 Submission of LaTeX articles

Authors who intend to submit a LaTeX article should obtain a copy of the NWS class file. This is available by anonymous FTP from

```
ftp.cup.cam.ac.uk
```

You will find the class file and instructions contained in a single file `nwscls.ltx` in the directory

```
pub/texarchive/journals/latex/nws-cls
```

The `readme.txt` (which is the same directory) tells you how to unpack the file `nwscls.ltx`. There may also be an '`unpacked`' directory containing all of the files separately, in case of difficulty. If you cannot obtain the NWS files, use the standard `article` class, with the default '`10pt`' option.

When submitting the final article, ensure that the following are included and are clearly labelled.

1. A hardcopy printout of the article.
2. The input file (exactly matching the hardcopy).
3. A copy of any user-defined macros.
4. If you have used BibTeX, the `.bib`, `.bbl` and `.bst` files that were used.
5. Any other files necessary to prepare the article for typesetting.

The files for the *final* article should be text-only with no system-dependent control codes.

2 Using the NWS class file

First, copy the file `nws.cls` (and `nws.bst` if you use BibTeX) into an appropriate subdirectory on your system. The NWS class is implemented as a complete document class, and *not* as an class option. In order to use the NWS class, replace `article` by `nws` in the `\documentclass` command at the beginning of your document: that is,

```
\documentclass{article}
```

is replaced by

```
\documentclass{nws}
```

Author-defined macros should be inserted before `\begin{document}`, or in a separate file and should be included with the submission. Authors must not change any of the macro definitions or parameters in `nws.cls`.

2.1 Document class options

In general, the following standard document class options should *not* be used with the NWS class file:

- `10pt`, `11pt` and `12pt` – unavailable;
- `twoside` is the default (`oneside` is disabled);
- `onecolumn` is the default (`twocolumn` is disabled);
- `titlepage` is not required and is disabled;
- `fleqn` and `leqno` should not be used, and are disabled.

The following new class options are provided:

- `prodtf` – tells the class file that we want to use the production typeface. This automatically sets the odd, even and top margins.

3 Additional facilities

In addition to all the standard LaTeX design elements, the NWS class includes the following features.

- Additional commands for typesetting the title page. Extended commands for specifying a short version of the title and author(s) for the running headlines.
- New options to the `\maketitle` command to create Functional and Theoretical Pearl(s).
- A `proof` environment.
- Control of enumerated lists.

Once you have used these additional facilities in your document, it can be processed only with `nws.cls`.

3.1 Titles, authors' names and running headlines

At the beginning of your article, the title should be generated in the usual way using the `\maketitle` command. Immediately following the title you may include an abstract and/or capsule review. For example, the titles for this guide were produced by the following source.

```
\title[Network Science]
      {\LaTeXe\ guide for Network Science}
\author[D.A. Tranah]
       {David Tranah\\
        Cambridge University Press, Cambridge CB2 2RU, UK\\
        \email{dtranah@cup.cam.ac.uk}}

\begin{document}
\maketitle

\begin{abstract}
This guide is for authors who are preparing papers...
\end{abstract}
```

In the NWS class, the title of the article and the author's name (or authors' names) are used both at the beginning of the article for the main title and throughout the article as running headlines at the top of every page. The title is used on odd-numbered pages (rectos) and the author's name appears on even-numbered pages (versos). The `\pagestyle` and `\thispagestyle` commands should *not* be used. Similarly, the commands `\markright` and `\markboth` should not be necessary.

Although the article title can run to several lines of text, the running headline must be a single line. Moreover, the title can incorporate new-line commands (e.g. `\\`), but these are not acceptable in a running headline. To enable you to specify an alternative short title, and an alternative short author's name, the standard `\title` and `\author` commands have been extended to take an optional argument to be used as the running headline.

```
\title[Short title]
      {Full title which can be as long as necessary}
\author[Author name]
       {AUTHOR NAME \\ Affiliation}
```

Notice that the author name in the argument for the running head should be in mixed case, and the author name for the title should be in upper case only. The author affiliation is set in the normal way, after a `\\` in the argument to the `\author` command.

Any 'work supported by' or 'authors current address' information should be inserted via `\thanks` commands, which should be positioned after the appropriate 'AUTHOR NAME' in the `\author` command.

If there are four (or more) authors for the article, the author running head should contain the first author name followed by 'et al.' only. e.g.

```
\author[Author1 et al.]
       {AUTHOR1...}
```

The previous examples show an article with one author, the normal LaTeX conventions have been extended to allow the author names and their affiliations to be typeset in the correct NWS style. The following examples should cover most possibilities:

Case 1. Two authors with the same affiliation:

```
\author[Author1 and Author2]
       {AUTHOR1 and AUTHOR2\\
        Affiliation for both authors}
```

If the author names are too long to fit onto one line, it should be broken into two or more lines using the `\authorbreak` command. Don't use `\\` to linebreak the author names – as this will not do what you expect.

Case 2. Two authors with different affiliations:

```
\author[Author1 and Author2]
       {AUTHOR1\\
        Affiliation for Author1
        \and AUTHOR2\\
        Affiliation for Author2}
```

Case 3. Three (or more) authors, two with the same affiliation:

```
\author[Author1, Author2 and Author3]
       {AUTHOR1, AUTHOR2\\
          Affiliation for Author1 and Author2
        \and AUTHOR3\\
          Affiliation for Author3}
```

3.2 Proofs

A new environment exists for creating Proofs, e.g. Use K_λ and S_λ to translate combinators into λ-terms. For the converse, translate $\lambda x \ldots$ by $[x] \ldots$ and use induction and the lemma. This was produced by the following code:

```
\begin{proof}
  Use $K_\lambda$ and $S_\lambda$ to...
\end{proof}
```

The end of proof marker is produced automatically. If you wish to omit this, use the `proof*` environment instead.

If a proof ends with a display equation, then it is customary for the proofbox to be positioned at the end of equation finishing the proof. Use K_λ and S_λ to translate combinators into λ-terms. For the converse, translate $\lambda x \ldots$ by $[x] \ldots$ and use induction and the lemma.

$$a_1 \equiv (2\Omega M^2/x)$$

Was produced with:

```
\begin{proof*}
  Use $K_\lambda$ and $S_\lambda$ to...
  \[ a_1 \equiv (2\Omega M^2/x) \mathproofbox \]
\end{proof*}
```

Notice the use of `proof*` to turn off the automatic proofbox.

The proof environment will also take an optional argument which allows you to produce 'special' proofs. e.g. [Proof of Theorem 27] We define a linear isometry $A : H \to H$ by the nonlinear Schrödinger equation. It would not be hard to modify the proof to obtain an analogous result for ellipsoids rather than spheres. Which was produced like this:

```
\begin{proof}[Proof of Theorem 27]
  We define a linear isometry...
\end{proof}
```

Notice that once the optional argument is used, you have to type all of the text which is to appear as the heading.

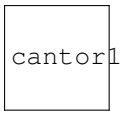


Figure 1: A Cantor repeller. Long figure captions will be indented left and right; short ones will be centred by default.

```
\begin{figure}
  \includegraphics[scale=0.55]{cantor1.eps}
  %  note that the square brace option below is only required
  %  if you intend to produce a list of illustrations
  \caption[Shortened figure caption for the list of illustrations]
    {A Cantor repeller. Long figure captions will be indented left
    and right; short ones will be centred by default.}
  \label{cantor}
\end{figure}
```

3.3 Figures

The NWS class will cope with most positioning of your figures. As captions fall below figures, the figure must be included first, then the caption, then the label. This is illustrated in Figure 1. The `cantor1.eps` file has been called in by using `\usepackage{graphicx}` in the preamble. Note that if you are producing a list of illustrations (using braces, but without the full point.

3.4 Programs

NWS encourages authors to use one of two styles for typesetting programs, *mathematical* and *verbatim*.

A program typeset in the mathematical style is shown in Figure 2, and the commands used to typeset this program are shown in Figure 3. This uses the ordinary mathematics mode of LaTeX: displayed programs are surrounded by `\[` and `\]`, and the `array` command is used for alignment. However, there are two important differences. First, the `\programmath` command appears before the program text; this causes math mode to use ordinary spacing for italic identifiers, rather than math spacing. The `\unprogrammath` command returns to normal math spacing. Second, in math mode spaces are ignored, so a tilde ~ is used instead. (In LaTeX, a tilde generates a "hard space" that is never replaced by a line break.) To include program text in mathematics style inline, surround it with dollar `$` signs. For example, the input

```
See how \programmath $differ~x$
differs from \unprogrammath $differ x$.
```

produces the output

See how $differ\ x$ differs from $differx$.

A program typeset in the verbatim style is shown in Figure 4, and the commands used to typeset this program are shown in Figure 5. This uses the ordinary verbatim mode of LaTeX: displayed programs are surrounded by `\begin{verbatim}` and `\end{verbatim}`, and alignment is indicated with spaces in the source file (don't use tabs, which may not be processed properly). To include program text in verbatim style inline, use the `\verb` command. For example, the input

```
On a terminal, this looks like \verb"differ x".
```

produces the output

On a terminal, this looks like `differ x`.

It is recommended that programs in figures be offset from the text using the `\figrule` command, as shown in Figures 2–5.

Some new macros have been provided for a few convenient symbols in math mode. These are illustrated in Table 1.

$$
\begin{array}{lcl}
exp & :: & Exp \to Arr \to Val \\
exp\ (Var\ i)\ a & = & index\ i\ a \\
exp\ (Const\ v)\ a & = & v \\
exp\ (Plus\ e_1\ e_2)\ a & = & exp\ e_1\ a + exp\ e_2\ a \\
 & & \\
com & :: & Com \to Arr \to Arr \\
com\ (Asgn\ i\ e)\ a & = & update\ i\ (exp\ e\ a)\ a \\
com\ (Seq\ c_1\ c_2)\ a & = & com\ c_2\ (com\ c_1\ a) \\
com\ (If\ e\ c_1\ c_2)\ a & = & \text{if } exp\ e\ a0 \text{ then } com\ c_1\ a \text{ else } com\ c_2\ a \\
 & & \\
prog & :: & Prog \to Val \\
prog\ (Prog\ c\ e) & = & exp\ e\ (com\ c\ (newarray\ 0))
\end{array}
$$

Figure 2: Example program in mathematical style.

```
\begin{figure}
\figrule
\programmath
\[
\begin{array}{lcl}
exp                    & :: & Exp \rightarrow Arr \rightarrow Val \\
exp~(Var~i)~a          & =  & index~i~a \\
exp~(Const~v)~a        & =  & v \\
exp~(Plus~e_1~e_2)~a   & =  & exp~e_1~a+exp~e_2~a \\
\\
com                    & :: & Com \rightarrow Arr \rightarrow Arr \\
com~(Asgn~i~e)~a       & =  & update~i~(exp~e~a)~a \\
com~(Seq~c_1~c_2)~a    & =  & com~c_2~(com~c_1~a) \\
com~(If~e~c_1~c_2)~a   & =  & \textrm{if } exp~e~a \dequals 0 \textrm{ then }
                                com~c_1~a \textrm{ else } com~c_2~a \\
\\
prog                   & :: & Prog \rightarrow Val\\
prog~(Prog~c~e)        & =  & exp~e~(com~c~(newarray~0))
\end{array}
\]
\unprogrammath
\caption{Example program in mathematical style.}\label{mathfigure}
\figrule
```

Figure 3: Typesetting the example program in mathematical style.

```
exp                 :: Exp -> Arr -> Val
exp (Var i) a       =  index i a
exp (Const v) a     =  v
exp (Plus e1 e2) a  =  exp e1 a + exp e2 a

com                 :: Com -> Arr -> Arr
com (Asgn i e) a    =  update i (exp e a) a
com (Seq c1 c2) a   =  com c2 (com c1 a)
com (If e c1 c2) a  =  if  exp e a == 0  then  com c1 a  else  com c2 a

prog                :: Prog -> Val
prog (Prog c e)     =  exp e (com c (newarray 0))
```

Figure 4: Example program in verbatim style.

```
\begin{figure}
\figrule
\begin{center}
\begin{verbatim}
exp                 :: Exp -> Arr -> Val
exp (Var i) a       =  index i a
exp (Const v) a     =  v
exp (Plus e1 e2) a  =  exp e1 a + exp e2 a

com                 :: Com -> Arr -> Arr
com (Asgn i e) a    =  update i (exp e a) a
com (Seq c1 c2) a   =  com c2 (com c1 a)
com (If e c1 c2) a  =  if  exp e a == 0  then  com c1 a  else  com c2 a

prog                :: Prog -> Val
prog (Prog c e)     =  exp e (com c (newarray 0))
\end{verbatim}
\end{center}
\caption{Example program in verbatim style.}\label{verbfigure}
\figrule
\end{figure}
```

Figure 5: Typesetting the example program in verbatim style.

Table 1: New symbol macros

Symbol	Usage	Keyed as
`\dplus`	$abc \dplus xyz$	`$abc \dplus xyz$`
`\dequals`	$abc \dequals xyz$	`$abc \dequals xyz$`
`\dcolon`	$abc \dcolon xyz$	`$abc \dcolon xyz$`
`\dcolonequals`	$abc \dcolonequals xyz$	`$abc \dcolonequals xyz$`

3.5 Abstract and Capsule reviews

The NWS class provides for an abstract and/or a capsule review; the abstract is produced by the following commands:

```
\begin{abstract}
     :
\end{abstract}
```

whereas the capsule review is produced by:

```
\begin{capsule}
     :
\end{capsule}
```

Either or both of these may be used, in either order, but it is assumed that, if both are used, there will be no other material between them. In NWS the abstract should precede any capsule review.

3.6 Lists

The NWS class provides the three standard list environments.

- Numbered lists, created using the `enumerate` environment;
- Bulleted lists, created using the `itemize` environment;
- Labelled lists, created using the `description` environment.

The `enumerate` environment numbers each list item with an arabic numeral; alternative styles can be achieved by inserting a redefinition of the number labelling command after the `\begin{enumerate}`. For example, a list numbered with roman numerals inside parentheses can be produced by the following commands:

```
\begin{enumerate}[(iii).]
  \renewcommand{\theenumi}{(\roman{enumi})}
  \item first item
        :
\end{enumerate}
```

This produces the following list:

1. first item
2. second item
3.

Notice that an optional argument "`(iii).`" has been given to the `enumerate` environment, specifying the *widest label* used in the list. This is because roman numerals are wider than the arabic numerals normally used by `enumerate`, and so the labels would otherwise have been pushed out into the margin.

4 User-defined macros

If you define your own macros, you must ensure that their names do not conflict with any existing macros in LaTeX (or if you are using this). You should also place them in the preamble of your input file, between the `\documentclass` (but after any `\usepackage` commands) and before the `\begin{document}` command.

Apart from scanning the indexes of the relevant manuals, you can check whether a macro name is already used by using `\newcommand`, which will check for the existence of the macro you are trying to define. If the macro exists LaTeX will respond with:

```
! LaTeX Error: Command ... already defined.
```

In this case you should choose another name, and try again.

Such macros must be in a place where they can easily be found and modified by the journal's editors or typesetter. They must be gathered together in the preamble of your input file, or in a separate `macros.tex` file with the command `% Content from macros would be included here"` in the preamble. Macro definitions must not be scattered about your document where they are likely to be completely overlooked by the typesetter.

The same applies to font definitions that are based on Computer Modern fonts. These must be changed by the typesetter to use the journal's correct typeface. In this case, you should draw attention to these font definitions on the hard copy that you submit for publication and by placing a comment in your input file just before the relevant definitions, for example `% replace font!`

5 Some guidelines for using standard facilities

The following notes may help you achieve the best effects with the NWS class file.

5.1 Sections

LaTeX provides five levels of section headings and they are all defined in the NWS class file:

Heading A – `\section{...}`

Heading B – `\subsection{...}`

Heading C – `\subsubsection{...}`

Heading D – `\paragraph{...}`

Heading E – `\subparagraph{...}`

Section numbers are given for sections, subsection and subsubsection headings.

5.2 Figures and tables

The `figure` and `table` environments are implemented as described in the LaTeX Manual to provide consecutively numbered floating inserts for illustrations and tables respectively. The standard inserts and their captions are formatted centred. Line breaks in captions can be inserted as required using `\\`.

5.2.1 Illustrations (or figures)

The NWS class will cope with most positioning of your illustrations and you should not normally use the optional positional qualifiers on the `figure` environment which would override these decisions. Figure captions should be below the figure itself, therefore the `\caption` command should appear after the figure or space left for an illustration.

Some figures in NWS will illustrate programs, as shown in Section 3.4 of this guide. Figure 6 shows an example of space left above a caption for artwork to be pasted in. This was produced with the following commands:

```
\begin{figure}
  \vspace{5cm} % the vertical depth of the artwork
  \caption{An example figure with space for artwork.}
  \label{sample-figure}
\end{figure}
```

Figure 6: An example figure with space for artwork.

The vertical depth should correspond roughly to the artwork you will submit; it will be adjusted to fit the final artwork exactly.

If your illustration extends over two pages, you can use the `\continuedfigure` facility. To use this, you key the figure caption for the second figure as follows:

```
\begin{figure}
  \continuedfigure
  \vspace{80pt}
  \caption{First figure, continued.}
  \label{continued}
\end{figure}
```

This ensures that the figure counter does not get incremented, and at the same time adds the word (cont.) to the caption. You may still use labels and references for this figure.

5.2.2 Tables

The NWS class file will cope with most positioning of your tables and you should not normally use the optional positional qualifiers on the `table` environment which would override these decisions. Normal journal style sets the table caption first, followed by a double rule, the table body and a double rule at the bottom. Single rules and spanner rules (`\cline`) can be used to separate headings from the columns. For example, Table 2 is produced using the following commands:

```
\begin{table}
  \caption{Results of Overloading for 3 Experimental Setups}
  \label{sample-table}
  \begin{minipage}{\textwidth}
    \begin{tabular}{lcrrrrr}
      \hline\hline
      Program& Expt.&
      CPU\footnote{Seconds of elapsed time on an unloaded Sun 3/50.}&
      RelCPU\footnote{CPU Time relative to experiment (a).}& GC&
      Mem\footnote{Bytes of heap used over the duration of the program.}&
      RelMem\footnote{Memory usage relative to experient (a).}\\
      \hline
      8 Queens& (a)&   2.88&  1.00&    6&   1.7M&  1.00\\
      &         (b)&  32.51& 11.29&  193&  48.9M& 28.76\\
      &         (c)&   7.90&  2.74&   42&  11.3M&  6.65\\
      \noalign{\vspace {.5cm}}
      Primes&   (a)&   4.89&  1.00&   19&   5.3M&  1.00\\
      &         (b)&  47.54&  9.72&  204&  54.5M& 10.28\\
      &         (c)&  10.08&  2.06&   47&  13.0M&  2.45\\
      \noalign{\vspace {.5cm}}
      Nfib&     (a)&  21.65&  1.00&  161&  40.4M&  1.00\\
      &         (b)& 221.65& 10.24& 1382& 349.0M&  8.64\\
      &         (c)&  21.30&  0.98&  161&  42.0M&  1.03\\
```

Table 2: Results of Overloading for 3 Experimental Setups

Program	Expt.	CPU[a]	RelCPU[b]	GC	Mem[c]	RelMem[d]
8 Queens	(a)	2.88	1.00	6	1.7M	1.00
	(b)	32.51	11.29	193	48.9M	28.76
	(c)	7.90	2.74	42	11.3M	6.65
Primes	(a)	4.89	1.00	19	5.3M	1.00
	(b)	47.54	9.72	204	54.5M	10.28
	(c)	10.08	2.06	47	13.0M	2.45
Nfib	(a)	21.65	1.00	161	40.4M	1.00
	(b)	221.65	10.24	1382	349.0M	8.64
	(c)	21.30	0.98	161	42.0M	1.03
KWIC	(a)	7.07	1.00	15	6.3M	1.00
	(b)	34.55	4.89	109	47.8M	7.59
	(c)	31.62	4.47	53	45.0M	7.14

[a]Seconds of elapsed time on a 62 unloaded Sun 4/50.
[b]CPU Time relative to experiment (a).
[c]Bytes of heap used over the duration of the program.
[d]Memory usage relative to experient (a).

```
    \noalign{\vspace {.5cm}}
    KWIC&      (a)&   7.07&  1.00&   15&   6.3M&  1.00\\
    &          (b)&  34.55&  4.89&  109&  47.8M&  7.59\\
    &          (c)&  31.62&  4.47&   53&  45.0M&  7.14\\
    \hline\hline
  \end{tabular}
  \vspace{-2\baselineskip}
 \end{minipage}
\end{table}
```

Notice the use of the '`\vspace{-2\baselineskip}`' command to remove the unwanted vertical space from above the table footnotes in this example.

Captions for 'continued' tables can be generated (in the same way as for figures) using the `\continuedtable` command. These should be positioned just before the `\caption` command in the appropriate table environment.

The `tabular` environment should be used to produce ruled tables; it has been modified for the NWS class in the following ways:

1. Additional vertical space is inserted above and below a horizontal rule (produced by `\hline`);
2. Tables are centred, and span the full width of the page; that is, they are similar to the tables that would be produced by `\begin{minipage}{\textwidth}`.

Because of this reformatting, vertical rules should not be used; furthermore, commands to redefine quantities such as `\arraystretch` should be omitted. If the old tabular facilities are needed, there is a new environment, `oldtabular`, which has none of the reformatting; it should be used in exactly the same way.

5.3 Appendices

You should use the standard LaTeX Appendices, normally, just before any references. From that point on `\section` will produce an appendix, which are numbered A, B etc., equations as (A1), (B1) etc. Figures and tables also number A 1, B 1 etc.

5.4 References

As with standard LaTeX, there are two ways of producing a list of references; either by using BibTeX with the NWS bibliography style `nws.bst`, or by compiling a list of references by hand (using a `thebibliography` environment).

5.4.1 Using BibTEX

If you have BibTEX installed on your system, the following is a brief description on how to automatically generate a bibliography (`.bbl` file) for your article. Your article should contain at least the following elements:

```
% sample.tex
\documentclass{nws}

\begin{document}
  \cite{citations}
  \verb"
```